Exercise : 2 **Roll No:240701581**

Develop and compare CLI, GUI, and Voice User Interfaces (VUI) for the same task and assess user satisfaction using Python (Tkinter for GUI, Speech Recognition for VUI), Terminal

AIM:

The aim is to develop and compare Command Line Interface (CLI), Graphical User Interface (GUI), and Voice User Interface (VUI) for the same task, and assess user satisfaction using Python (with Tkinter for GUI and Speech Recognition for VUI) and Terminal.

PROCEDURE:

i) CLI (Command Line Interface)

CLI implementation where users can add, view, and remove tasks using the terminal.

CODE :

```
# CLI - Library Book Manager

books = [ ]

def menu():

    print("\n====== LIBRARY BOOK SYSTEM ======")
    print("1. Add Book")
    print("2. Display Books")
    print("3. Remove Book")
    print("4. Exit")

while True:
    menu()
    option = input("Choose an option: ")

    if option == "1":
        name = input("Enter book name: ")
        books.append(name)
```

```
        print("Book successfully added to library.")

    elif option == "2":

        if not books:
            print("Library is empty.")
        else:
            print("\nAvailable Books:")
            for i, b in enumerate(books, start=1):
                print(f"{i}. {b}")

    elif option == "3":

        if not books:
            print("No books to remove.")

        else:
            for i, b in enumerate(books, start=1):
                print(f"{i}. {b}")
            num = int(input("Enter book number to remove: "))
            if 1 <= num <= len(books):
                removed = books.pop(num - 1)
                print(f"{removed} removed from library.")
            else:
                print("Invalid selection.")

    elif option == "4":

        print("Closing Library System.")
        break

    else:
        print("Wrong option selected.")
```

SAMPLE OUTPUT:

```
...
    ====== LIBRARY BOOK SYSTEM ======
    1. Add Book
    2. Display Books
    3. Remove Book
    4. Exit
    Choose an option: 1
    Enter book name: USER INTERFACE DESIGN
    Book successfully added to library.

    ====== LIBRARY BOOK SYSTEM ======
    1. Add Book
    2. Display Books
    3. Remove Book
    4. Exit
    Choose an option: 2

    Available Books:
    1. USER INTERFACE DESIGN

    ====== LIBRARY BOOK SYSTEM ======
    1. Add Book
    2. Display Books
    3. Remove Book
    4. Exit
    Choose an option: 3
    1. USER INTERFACE DESIGN
    Enter book number to remove: 1
    USER INTERFACE DESIGN  removed from library.

    ====== LIBRARY BOOK SYSTEM ======
    1. Add Book
    2. Display Books
    3. Remove Book
    4. Exit
    Choose an option: 4
    Closing Library System.
```

ii) GUI (Graphical User Interface)

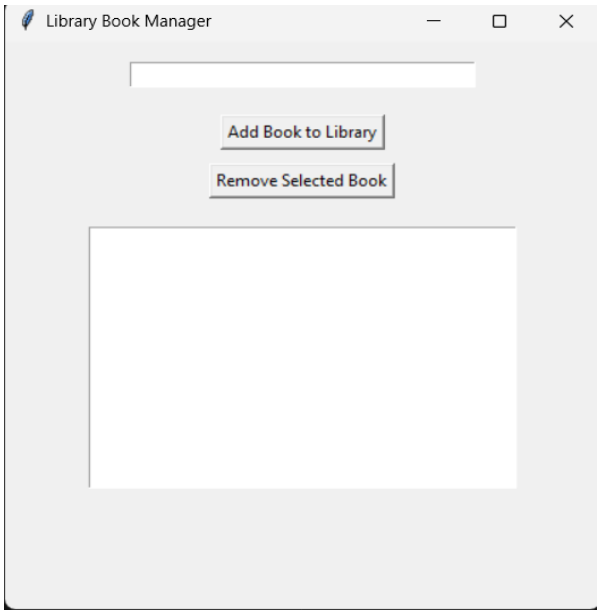
Tkinter to create a simple GUI for our To-Do List application.

CODE :

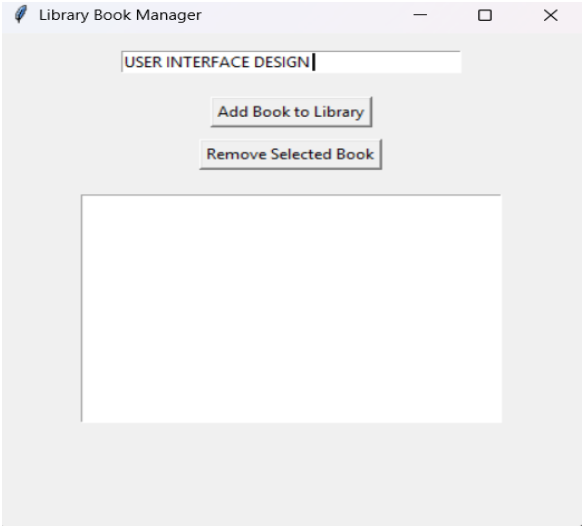
```
import tkinter as tk
from tkinter import messagebox

library = [ ]

def add_book():

    book = entry.get()
    if book.strip() == "":
        messagebox.showerror("Error", "Book name cannot be empty.")

    else:
        library.append(book)
        entry.delete(0, tk.END)
        refresh_list()

def refresh_list():
    book_list.delete(0, tk.END)
    for item in library:
        book_list.insert(tk.END, item)

def remove_book():

    try:
        selected_index = book_list.curselection()[0]
        deleted_book = library.pop(selected_index)
        refresh_list()
        messagebox.showinfo("Removed", f"{deleted_book} deleted.")
    except:
        messagebox.showwarning("Warning", "Please select a book.")

window = tk.Tk()
window.title("Library Book Manager")
window.geometry("420x420")

entry = tk.Entry(window, width=40)
entry.pack(pady=15)
```

```
add_btn = tk.Button(window, text="Add Book to Library", command=add_book)
add_btn.pack(pady=5)

remove_btn = tk.Button(window, text="Remove Selected Book", command=remove_book)
remove_btn.pack(pady=5)

book_list = tk.Listbox(window, width=50, height=12)
book_list.pack(pady=15)

window.mainloop()
```

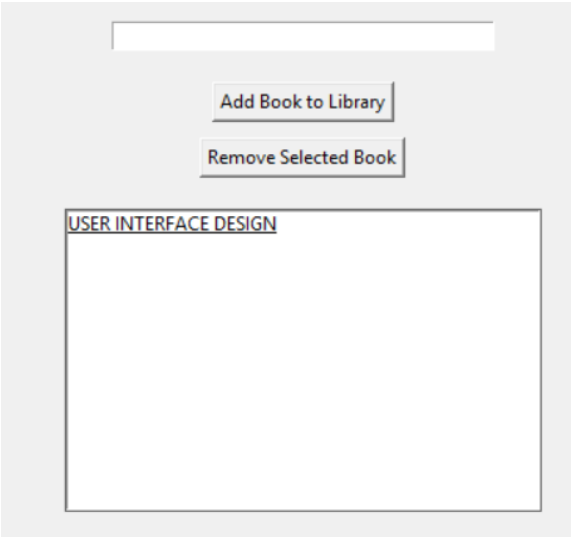
SAMPLE OUTPUT :



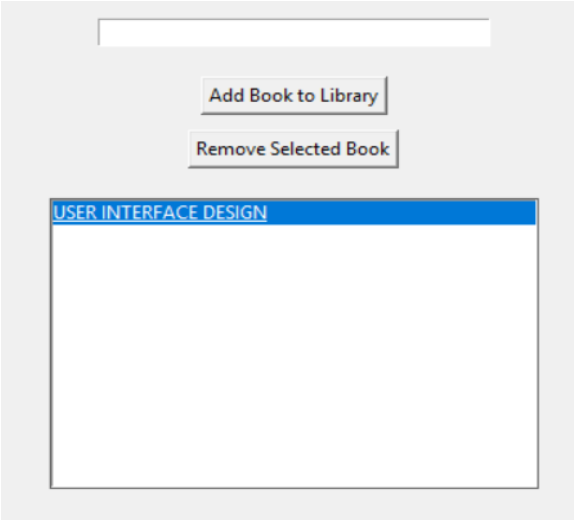
OPTION 1 : TO ADD BOOK (ENTER BOOK NAME AND CLICK ADD BOOK)



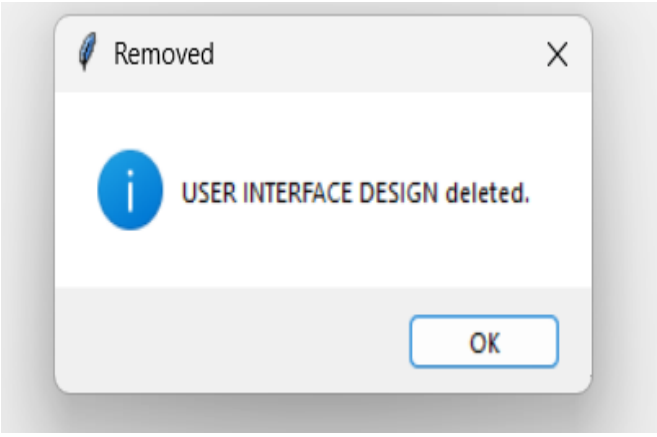
OUTPUT :



OPTION 2 : TO REMOVE THE BOOK(SELECT AND CLICK REMOVE BOOK)



OUTPUT :



i) VUI (Voice User Interface)

speech_recognition library for voice input and the pyttsx3 library for text-to-speech output. Make sure you have these libraries installed (pip install SpeechRecognition pyttsx3).

CODE :

```
import os
import platform
import speech_recognition as sr

recognizer = sr.Recognizer()

def listen():
    with sr.Microphone() as source:
        print(" Say 'system info'")
        audio = recognizer.listen(source)

    try:
        command = recognizer.recognize_google(audio)
        return command.lower()
    except:
        return ""

def show_info():
    print("Operating System:", platform.system())
    print("OS Version:", platform.version())
    print("Current User:", os.getlogin())

if __name__ == "__main__":
    command = listen()
    if "system" in command:
        show_info()
    else:
        print("✕ Command not recognized")
```

OUTPUT :

```
> Say 'system info'
O Operating System: Windows
O OS Version: 10.0.26100
O Current User: VELVIZHI

PS C:\Users\VELVIZHI\OneDrive\Deskto
p\New folder\New folder (2)
```